Ch4. Integrals

§4.1 Areas and Distances

[The Area Problem] - the area under a curve

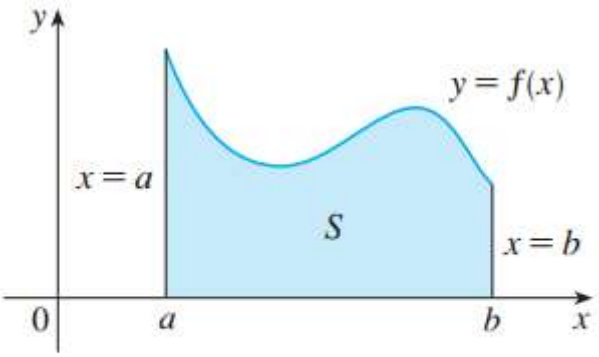
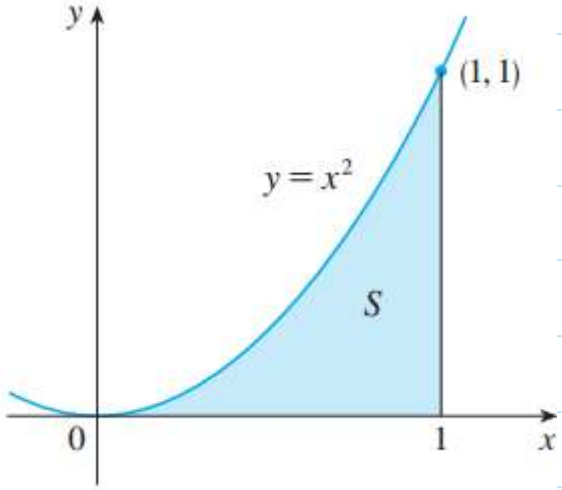
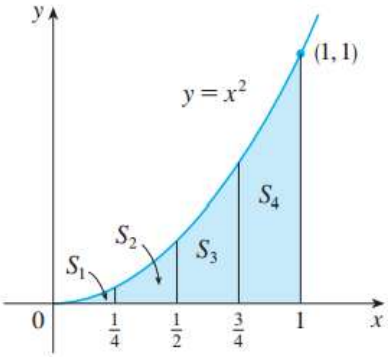


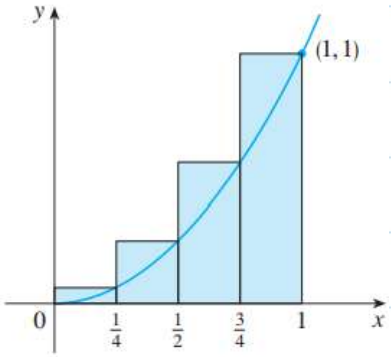
FIGURE 1

$S = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}$





(a)



(b)

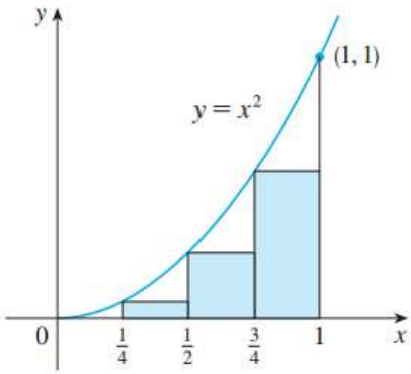
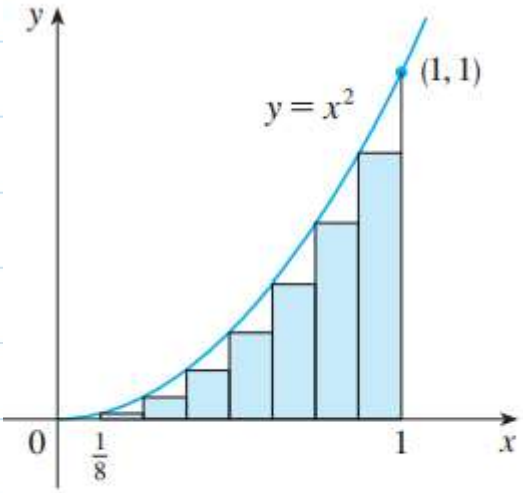
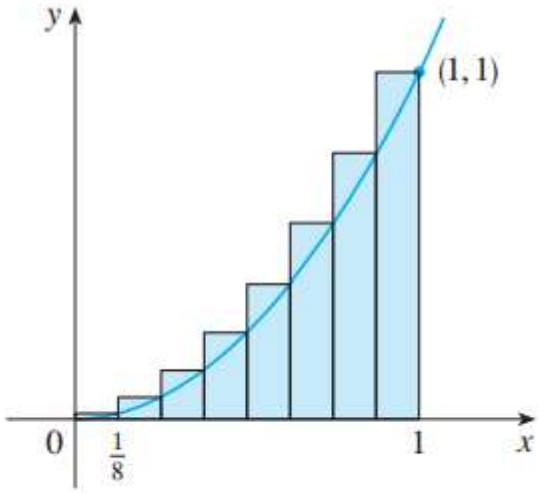


FIGURE 5



(a) Using left endpoints



(b) Using right endpoints

Example For the region S, show that the sum of areas of the upper approximating rectangles approaches $\frac{1}{3}$, that is, $\lim_{n\to\infty} R_n = \frac{1}{3}$.

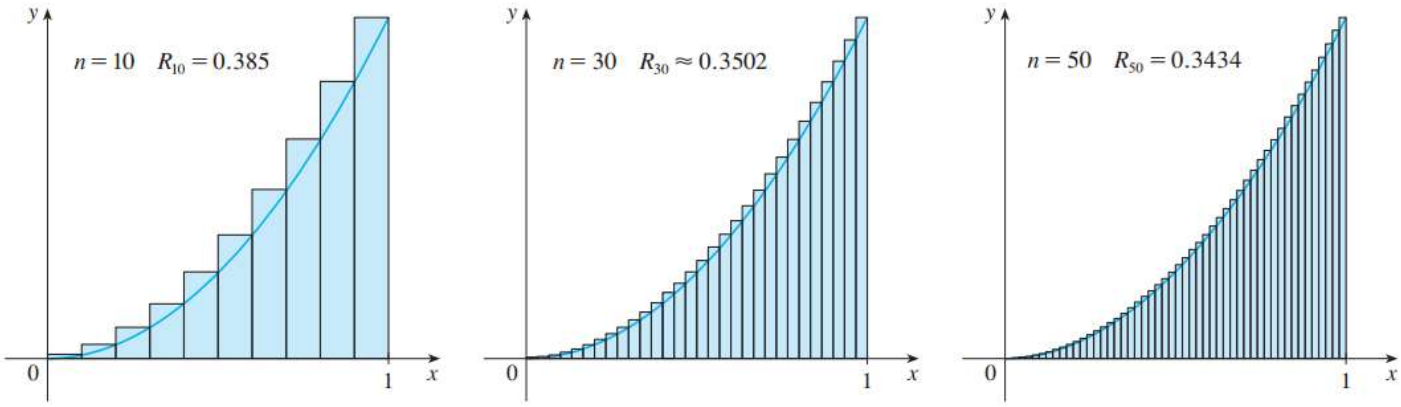


FIGURE 8 Right endpoints produce upper sums because $f(x) = x^2$ is increasing.

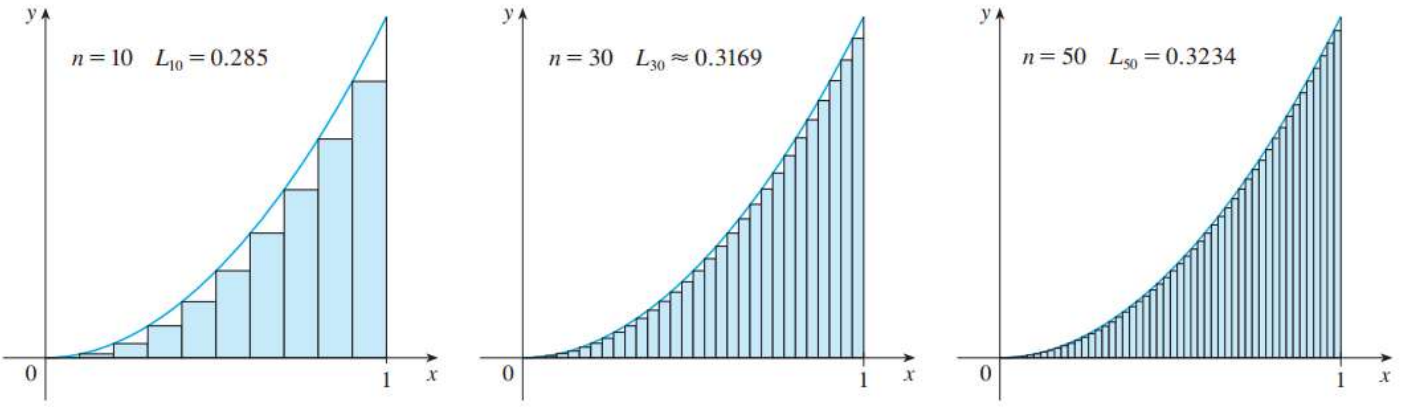
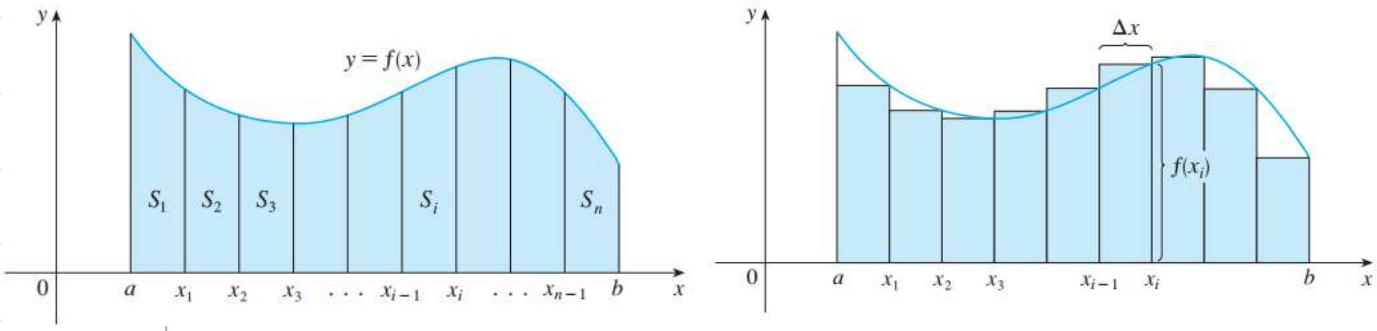


FIGURE 9 Left endpoints produce lower sums because $f(x) = x^2$ is increasing.



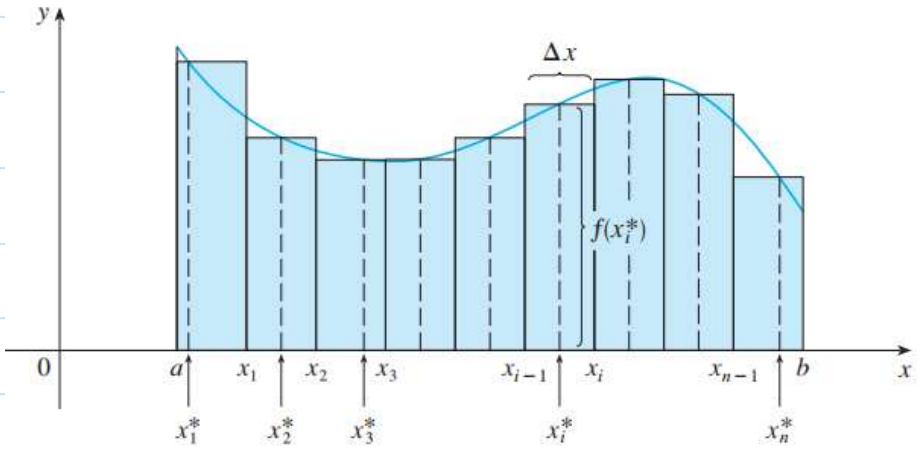
Divide $[a, b]$ into n subintervals $[x_i, x_{i+1}]$ ($0 \le i \le n-1$) with $a = x_0$, $b = x_n$.

Then $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\Delta x$.

Definition The area A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles :

$$A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} [f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x]$$

§4.2 The Definite Integral

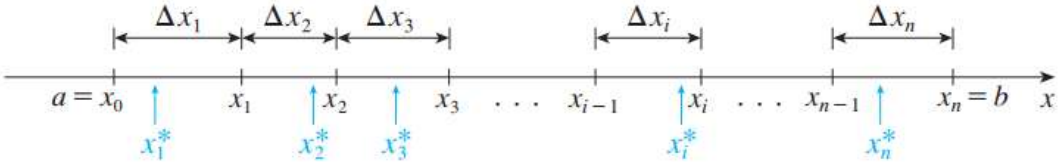


$$A = \lim_{n\to\infty}[f(x_1^*)\Delta x_1 + f(x_2^*)\Delta x_2 + \cdots + f(x_n^*)\Delta x_n]$$
$$= \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x_i$$

Let f be a continuous function defined on $[a,b]$. Divide $[a,b]$ into n subintervals by choosing partition points $a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$.

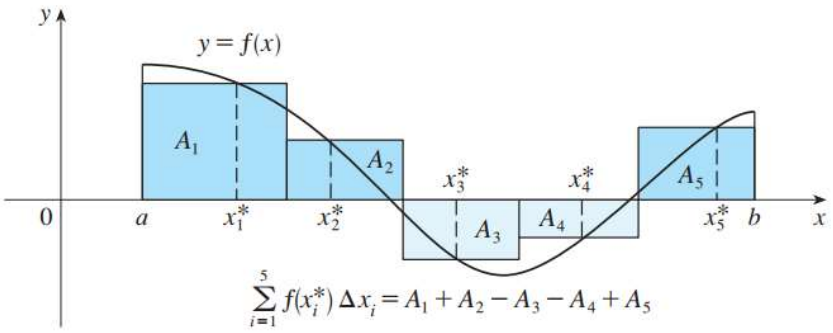
Then $\Delta x_i = x_i - x_{i-1}$: the length of the ith subinterval.

Let $x_i^* \in [x_{i-1}, x_i]$ be a sample point from the ith subinterval.



$$\sum_{i=1}^{n} f(x_i^*)\Delta x_i$$

: a Riemann sum



Definition [Definite integral]

If f is a function defined on $[a,b]$, the **definite integral of f from a to b** is the number

$$\int_a^b f(x)dx = \lim_{\max \triangle x_i \to 0} \sum_{i=1}^{n} f(x_i^*) \triangle x_i$$

provided that the limit exists. If it does exist, we say that f is **integrable** on $[a,b]$.

$$\int_a^b f(x)dx$$

Note $\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$

Theorem If f is continuous on $[a,b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a,b]$; that is, the definite integral $\int_a^b f(x)dx$ exists.

Theorem If f is integrable on $[a,b]$, then $\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i) \triangle x$,

where $\triangle x = \dfrac{b-a}{n}$ and $x_i = a + i\triangle x$.

Example (a) Evaluate the Riemann sum for $f(x) = x^3 - 6x$ taking the sample points to be right endpoints and $a=0,\ b=3,\ n=6$.

(b) Evaluate $\int_0^3 x^3 - 6x\ dx$

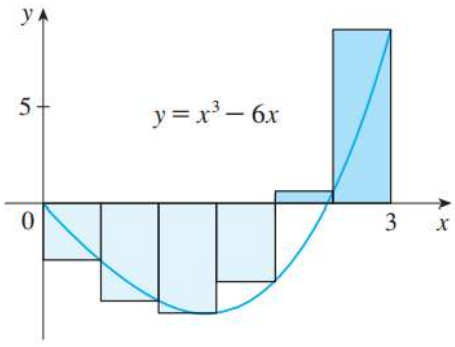


FIGURE 7

Example Evaluate the following integrals by interpreting each in terms of areas.

(a) $\int_0^1 \sqrt{1-x^2}\,dx$ (b) $\int_0^3 x-1 dx$

[Properties of the **Definite Integral**]

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

$$\int_a^a f(x)\,dx = 0$$

PROPERTIES OF THE INTEGRAL Suppose all the following integrals exist.

1. $\int_a^b c\,dx = c(b-a)$, where c is any constant
2. $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$
3. $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$, where c is any constant
4. $\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

5. $$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$$

COMPARISON PROPERTIES OF THE INTEGRAL

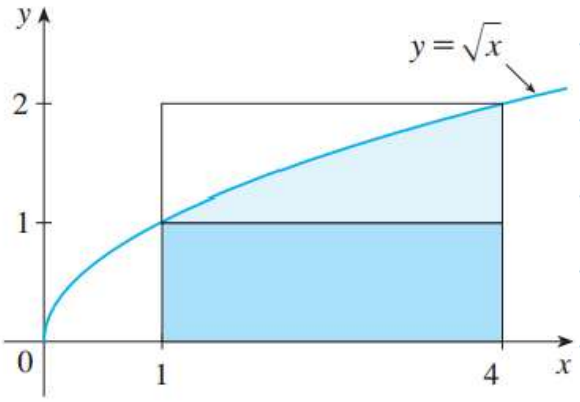
6. If $f(x) \geqslant 0$ for $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx \geqslant 0$.
7. If $f(x) \geqslant g(x)$ for $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx \geqslant \int_a^b g(x)\,dx$.
8. If $m \leqslant f(x) \leqslant M$ for $a \leqslant x \leqslant b$, then

$$m(b-a) \leqslant \int_a^b f(x)\,dx \leqslant M(b-a)$$

Example Evaluate $\int_0^1 4+3x^2\,dx$ if $\int_0^1 x^2 dx = \frac{1}{3}$.

Example If it is known that $\int_0^{10} f(x)dx = 17$ and $\int_0^{7} f(x)dx = 12$, find $\int_7^{10} f(x)dx$.

Example Estimate $\int_1^4 \sqrt{x}\,dx$.



§4.3 Evaluating Definite Integrals

Theorem [Evaluation Theorem]

If f is continuous on the interval $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a),$$

where F is any antiderivative of f, that is, $F' = f$.

why)

Notation $\int_a^b f(x)dx = F(x)|_a^b = F(x)\,]_a^b = [F(x)\,]_a^b = F(b) - F(a)$

Example Evaluate $\int_{-2}^{1} x^3 dx$.

Ch4. Integrals

[Indefinite Integrals]

Note $\int f(x)dx = F(x)$ means $F'(x) = f(x)$.

A definite integral $\int_a^b f(x)dx$

An indefinite integral $\int f(x)dx$

1 TABLE OF INDEFINITE INTEGRALS

$$\int cf(x)\,dx = c\int f(x)\,dx \qquad \int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int k\,dx = kx + C \qquad \int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \csc^2 x\,dx = -\cot x + C$$

$$\int \sec x \tan x\,dx = \sec x + C \qquad \int \csc x \cot x\,dx = -\csc x + C$$

Example Find the general indefinite integral $\int 10x^4 - 2\sec^2 x dx$.

Example Evaluate $\int_0^3 x^3 - 6x dx$.

Example Find $\int_0^{12} x - 12\sin x dx$.

Example A particle moves along a line so that its velocity at time t is $v(t) = t^2 - t - 6$ (measured in meters per second).

(a) Find the displacement of the particle during the time period $1 \leq t \leq 4$.

(b) Find the distance traveled during the time period $1 \leq t \leq 4$.

Ch4. Integrals